Block practice · Computational models, the CPU and memory

Work through each question type in order. Each section has a worked-example retry and all further practice. Attempt only the part outlined in red.

Input, process, output and other computational models

RAM, ROM, virtual memory and flash memory

The stored program concept and the parts of the CPU

Buses and the fetch-decode-execute cycle

What makes a CPU faster: clock speed, cores and cache

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Practice workbook · Input, process, output and other computational models

Naming the three stages of the IPO model or pulling an input, a process and an output out of a described program or problem; recognising sequential, parallel and multi-agent models; splitting a calculation across parallel processors.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-01

1. Question 2 · part b.ii · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b.ii only.

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2 Programs can process numerical data.

(a) Identify the method used to check input data matches set rules.

(1)

- ☒ **A** Abstraction
- ☒ **B** Encryption
- ☒ **C** Iteration
- ☒ **D** Validation

(b) **Figure 2** shows a section from an employee payslip produced by a computer program.

Pay Rate	Hours		Total Pay
£ 18.75	19		£ 356.25
	Tax	20%	£ 71.25
		Income	£ 285.00

Figure 2

(i) Complete the table to give **one** data type shown in **Figure 2** and give an example value.

(2)

Data type	
Example	

(ii) Complete the table to give an input, a process and an output used in the program to produce the payslip.

(3)

Input	
Process	
Output	

1. Question 2 · part b.ii · 3 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b.ii only.

(c) Numerical values can be represented in denary, binary and hexadecimal.

A program converts denary values in the range 0 to 255 into hexadecimal.

(i) The program is tested using appropriate integer values.

Complete the table to show an example of normal, boundary and erroneous test data for this program.

(2)

	Test data
Normal	
Boundary	
Erroneous	

(ii) The program uses subprograms to do the conversion.

Open **Q02cii** in the code editor.

Amend the code to complete the program to:

- convert the user input to integer
- check the input is between 0 and 255
- call the **getHex** subprogram, with the user input as an argument.

Do not add any further functionality.

Save your code as **Q02ciiFINISHED** with the correct file extension for the programming language.

(4)

(Total for Question 2 = 12 marks)

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1. Question 2 · part b.ii · 3 marks · Working and answer

Working and answer

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2. Question 1 · part a · 3 marks

2019 JUN · Paper 01

Further practice · Attempt part a only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

.....

2 .....

.....

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2. Question 1 · part a · 3 marks · Working and answer

Working and answer

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3. Question 3 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

3 The central processing unit (CPU) and other hardware components work together with the operating system to run programs and store data.

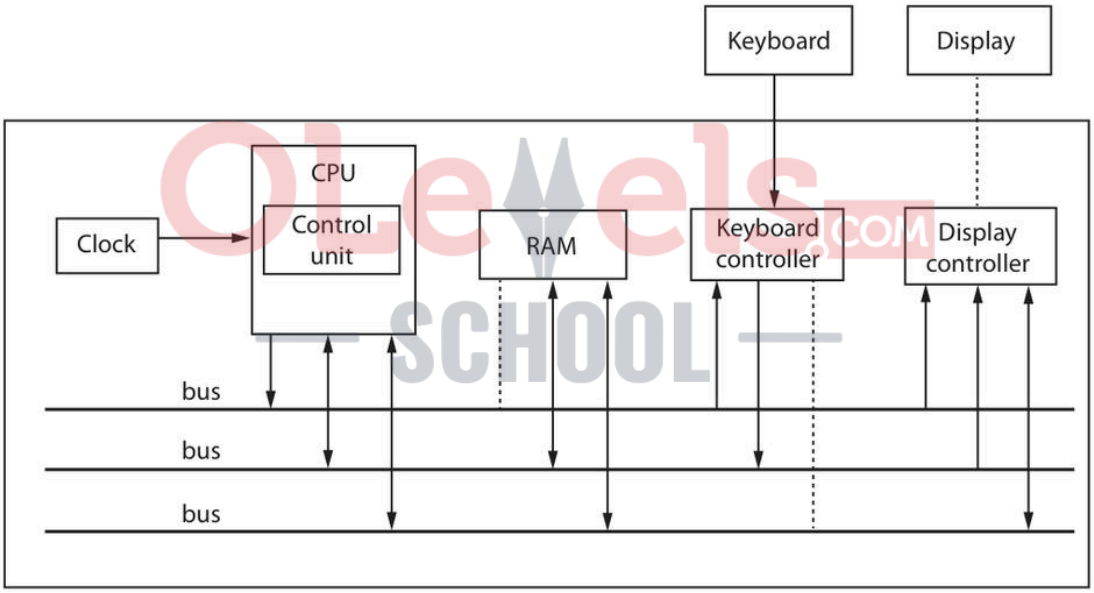
(a) Here is a partially completed diagram showing the hardware components used in the fetch-decode-execute cycle.

The function of the I/O controller is to manage communication between the CPU and input/output devices.

Complete the diagram by:

- labelling the buses
- adding directional arrows to the dotted lines to show the flow of communication.

(6)



(b) The CPU includes an area of cache memory.

Increasing the size of the cache will improve the performance of the CPU.

Explain how increasing the size of the cache improves the performance.

(2)

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3. Question 3 · part c · 4 marks

2020 NOV · Paper 01

Further practice · Attempt part c only.

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(c) Some computers have more than one CPU. The CPUs work in parallel to execute instructions.

Complete the table to show how two CPUs working in parallel could execute the expression:

(5 * 2) + (4 * 3) – (24 / 4)

(4)

CPU 1	CPU 2
5 * 2 = 10	

(d) One function of an operating system is to manage files on secondary storage.

Describe how an operating system manages the storage of a file on random access secondary storage.

(4)

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(Total for Question 3 = 16 marks)

3. Question 3 · part c · 4 marks · Working and answer

Working and answer

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4. Question 1 · part d · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part d only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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4. Question 1 · part d · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part d only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

.....

.....

(ii) State the purpose of an assembler.

(1)

.....

(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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4. Question 1 · part d · 1 mark · Working and answer

Working and answer

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5. Question 5 · part a · 2 marks

2023 JUN · Paper 01

Further practice · Attempt part a only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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5. Question 5 · part a · 2 marks

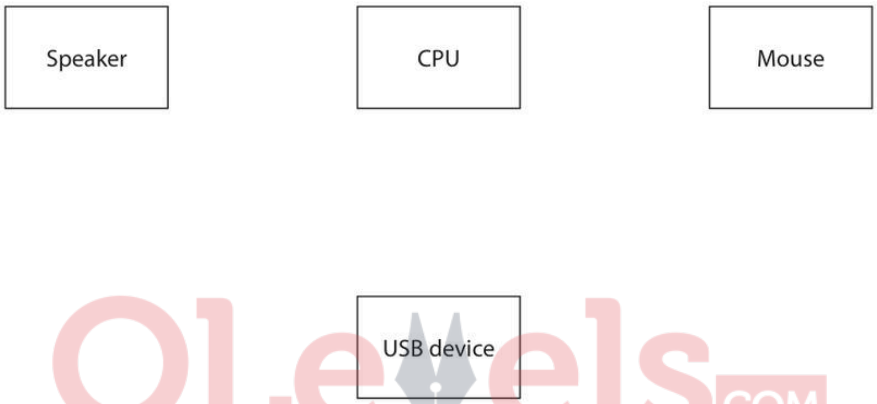
2023 JUN · Paper 01

Further practice · Attempt part a only.

(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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5. Question 5 · part a · 2 marks · Working and answer

Working and answer

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6. Question 1 · part a · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

1 Programs are used to handle financial transactions.

(a) Monthly account statements are created by a program.

Figure 1 shows a statement.

Date	Description	Debit (£)	Credit (£)	Balance (£)
01 Jan	Opening balance			128.35
04 Jan	Music store	10.23		118.12
07 Jan	Salary		1,515.28	1,633.40
11 Jan	Electricity company	50.00		1,583.40
19 Jan	Uniform shop	12.56		1,570.84
27 Jan	Refund		25.00	1,595.84
29 Jan	Rent	750.00		845.84
31 Jan	Interest		0.25	846.09

Figure 1

Complete the table to identify an input, an output and a process used by the program to generate the **furthest right column**.

(3)

Input	
Processing	
Output	

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6. Question 1 · part a · 3 marks

2023 JUN · Paper 02

Further practice · Attempt part a only.

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(b) A tax rate is applied to a gross value to give a net value.

Open **Q01b** in the code editor.

Use the code to answer these questions.

(i) Give the contents of a comment.

(1)

(ii) Identify a logical operator used in this program.

(1)

(iii) Give the name of a global variable.

(1)

(iv) Give the name of a constant.

(1)

(v) Give the name of a parameter.

(1)

(c) Variables and constants are used in program code.

(i) State the purpose of a constant.

(1)

(ii) Give **one** benefit of using a constant.

(1)

(Total for Question 1 = 10 marks)

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6. Question 1 · part a · 3 marks · Working and answer

Working and answer

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7. Question 4 · part b · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part b only.

4 Peter and Marie are employees at a research facility.

(a) Peter is creating a predator-prey population simulation of rabbits and foxes.

He creates rules to simulate the movement and feeding patterns for each animal over time.

One benefit of the predator-prey simulation is that it is simpler than monitoring real life animal populations.

Explain **one other** benefit of using this simulation.

(2)

(b) Identify the computational model in which several computer systems work together to achieve a goal.

(1)

- ☒ **A** Sequential model
- ☒ **B** Parallel model
- ☒ **C** Multi-agent model
- ☒ **D** Input-process-output model

(c) Employees at the research facility use a biometric identifier to login to the network.

Describe **one other** security measure that could be used to ensure that an employee is who they claim to be.

(2)

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7. Question 4 · part b · 1 mark · Working and answer

Working and answer

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Practice workbook · RAM, ROM, virtual memory and flash memory

The function and properties of RAM, ROM and virtual memory, why a computer needs both RAM and ROM, what is stored in ROM, and how flash differs from RAM.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-05

1. Question 2 · part b.i · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

2 Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

(1)

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

(3)

1 .....

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2 .....

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3 .....

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1. Question 2 · part b.i · 2 marks

2024 JUN · Paper 01

Worked-example retry · Attempt part b.i only.

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(b) Computers use different types of storage.

(i) Explain **one** reason why computers need both RAM and ROM.

(2)

(ii) Describe how data is read from a magnetic hard drive.

(3)

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(c) Computers perform the fetch-decode-execute cycle.

Complete the table by writing the numbers 1 to 4 in the sequence number column to put each stage in the correct order.

Stage	Sequence number
The instruction is placed onto the data bus	
The Arithmetic and Logic Unit (ALU) performs a calculation	
The address of the next instruction is placed onto the address bus	
The instruction is decoded by the control unit	

(4)

1. Question 2 · part b.i · 2 marks · Working and answer

Working and answer

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2. Question 1 · part d · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

.....

2 .....

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2. Question 1 · part d · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part d only.

(d) Some computers use virtual memory.

Explain how virtual memory works.

(2)

(e) An optical storage device writes data onto a CD or DVD.

Describe how data is stored physically on optical media.

(2)

(f) A disc-formatting program is an example of a type of software.

Identify the type.

(1)

- ☒ **A** Security software
- ☒ **B** Application software
- ☒ **C** Translation software
- ☒ **D** System software

(Total for Question 1 = 11 marks)

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2. Question 1 · part d · 2 marks · Working and answer

Working and answer

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3. Question 3 · part b.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part b.ii only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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3. Question 3 · part b.ii · 2 marks · Working and answer

Working and answer

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4. Question 5 · part d.iii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.iii only.

5 Viza Health Centre is located in the North East of England.

(a) The health centre uses artificial intelligence to provide a symptom-checking service for its patients.

Patients log on to the website and input their symptoms.

(i) Describe how artificial intelligence could identify what is wrong with them.

(2)

(ii) Give **one** reason why a patient may not want to use this online service.

(1)

(b) The health centre has clinics in two buildings: Cleveland and Stockton.

The network server is in the Cleveland building.

(i) Name the type of network used to access the server from within the Cleveland building.

(1)

(ii) Name the type of network used to access the server from the Stockton building.

(1)

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P 7 2 5 3 8 A 0 1 6 2 0

4. Question 5 · part d.iii · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part d.iii only.

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(c) The network is at risk of an eavesdropping attack.

Identify the description of eavesdropping.

(1)

- ☒ **A** Tricking people into giving information by sending emails pretending to be from someone in authority
- ☒ **B** Spying on someone using a computer
- ☒ **C** Intercepting information as it is transmitted over a network
- ☒ **D** Redirecting a user from a genuine website to a fake one

(d) Doctors use laptops when they visit patients in their homes.

(i) The laptops have solid state drives.

Explain **one** reason why a solid state drive is better than a magnetic hard drive for the laptops.

(2)

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(ii) Describe how data is stored on a solid state drive.

(2)

(iii) The laptops have two types of memory.

Complete the table by adding **one** tick (✓) to match **each** description to the type of memory used.

(2)

Description	RAM	ROM
Stores the boot up sequence		
The contents are lost when the laptop is shut down		

(Total for Question 5 = 12 marks)

4. Question 5 · part d.iii · 2 marks · Working and answer

Working and answer

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5. Question 5 · part b.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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5. Question 5 · part b.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

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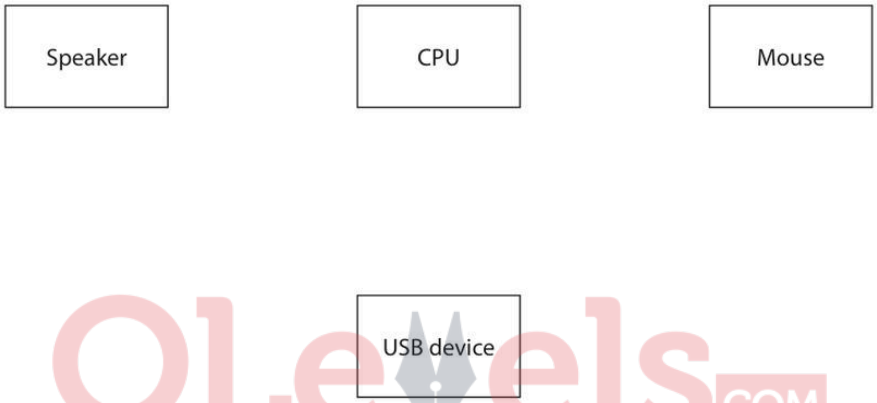
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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5. Question 5 · part b.ii · 1 mark

2023 JUN · Paper 01

Further practice · Attempt part b.ii only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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5. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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6. Question 2 · part b.i · 1 mark

2025 JUN · Paper 01

Further practice · Attempt part b.i only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

.....

2 .....

.....

(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

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(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

.....

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.....

P 7 8 6 3 7 A 0 5 2 0

6. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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7. Question 4 · part c.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c.ii only.

4 A house has solar panels and a battery storage system.

(a) A smartphone application is used to monitor the battery storage system remotely. The smartphone uses a 5G connection.

(i) Give **two** benefits of a 5G connection.

(2)

1 .....

.....

2 .....

.....

(ii) The application requires users to authenticate their identity.

One method of authentication is a user ID and password.

Describe **one other** method the application could use.

(2)

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(iii) Software developers perform code reviews.

Give **two** reasons for carrying out a code review.

(2)

1 .....

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2 .....

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7. Question 4 · part c.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c.ii only.

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(b) The battery storage system stores readings as signed 8-bit binary integers in **sign and magnitude** format.

Complete the table to show the 8-bit binary pattern and denary value for the lowest and the highest values that can be stored.

(2)

	8-bit binary pattern	Denary value
Lowest value		
Highest value		

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7. Question 4 · part c.ii · 2 marks

2025 JUN · Paper 01

Further practice · Attempt part c.ii only.

(c) The solar panels and battery storage system are controlled by a program.

(i) An LED is displayed when the system is on (**O**), the sun is shining (**S**), and the house is not using all the power (**P**) generated by the solar panels.

Complete the truth table to show the results of each operation.

(3)

O	S	P	O AND S	NOT P	(O AND S) AND (NOT P)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

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(ii) The start-up instructions for the solar panels and battery storage system are stored in read-only memory (ROM).

Explain why the start-up instructions are stored in ROM.

(2)

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(Total for Question 4 = 13 marks)

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7. Question 4 · part c.ii · 2 marks · Working and answer

Working and answer

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Practice workbook · The stored program concept and the parts of the CPU

Von Neumann stored program idea, what instructions and memory addresses are, and the roles of the control unit, ALU and registers, including labelling them on a diagram.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-03

1. Question 1 · part a.i · 1 mark

2022 JUN · Paper 01

Worked-example retry · Attempt part a.i only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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2

P 7 2 5 3 8 A 0 2 2 0

1. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

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2. Question 1 · part b · 1 mark

2019 JUN · Paper 01

Further practice · Attempt part b only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
- ☒ **B** Central processing unit
- ☒ **C** ROM
- ☒ **D** Address bus

(c) Von Neumann developed the stored program concept that permits two different types of item to reside in memory.

Name these **two** types.

(2)

1 .....

.....

2 .....

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2. Question 1 · part b · 1 mark · Working and answer

Working and answer

OLevels.com SCHOOL

3. Question 1 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

1 Computers are made up of both hardware and software components.

(a) Computers use a three-step computational model.

Complete the diagram with the names of the **three** steps.

(3)



(b) Identify the hardware component responsible for controlling the fetch-decode-execute cycle.

(1)

- ☒ **A** Cache
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Name these **two** types.

(2)

1 .....

.....

2 .....

.....

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3. Question 1 · part c · 2 marks

2019 JUN · Paper 01

Further practice · Attempt part c only.

(d) Some computers use virtual memory.

Explain how virtual memory works.

(2)

(e) An optical storage device writes data onto a CD or DVD.

Describe how data is stored physically on optical media.

(2)

(f) A disc-formatting program is an example of a type of software.

Identify the type.

(1)

- ☒ **A** Security software
- ☒ **B** Application software
- ☒ **C** Translation software
- ☒ **D** System software

(Total for Question 1 = 11 marks)

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3. Question 1 · part c · 2 marks · Working and answer

Working and answer

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4. Question 4 · part d.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.i only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

(1)

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

(2)

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

(3)

R	S	W	NOT S	R AND W	(NOT S) OR (R AND W)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 1 0 2 0

4. Question 4 · part d.i · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.i only.

DO NOT WRITE IN THIS AREA

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DO NOT WRITE IN THIS AREA

(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

(2)

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

(2)

1 .....

2 .....

(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

(2)

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

(2)

Change made	Effect
Increasing the width of the data bus	
Increasing the width of the address bus	

(Total for Question 4 = 14 marks)

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4. Question 4 · part d.i · 2 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 4 · part d.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.ii only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

(1)

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

(2)

.....

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(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

(3)

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0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

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5. Question 4 · part d.ii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.ii only.

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

(2)

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

(2)

1 .....

2 .....

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(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

(2)

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

(2)

Change made	Effect
Increasing the width of the data bus	
Increasing the width of the address bus	

(Total for Question 4 = 14 marks)

5. Question 4 · part d.ii · 2 marks · Working and answer

Working and answer

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6. Question 5 · part a · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

5 Computers can be general-purpose or specific-purpose machines.

(a) The central processing unit (CPU) is responsible for executing a program.

Figure 4 shows some of the hardware components of a computer system.

Complete the table by matching the component to the letters shown in **Figure 4**.

(5)

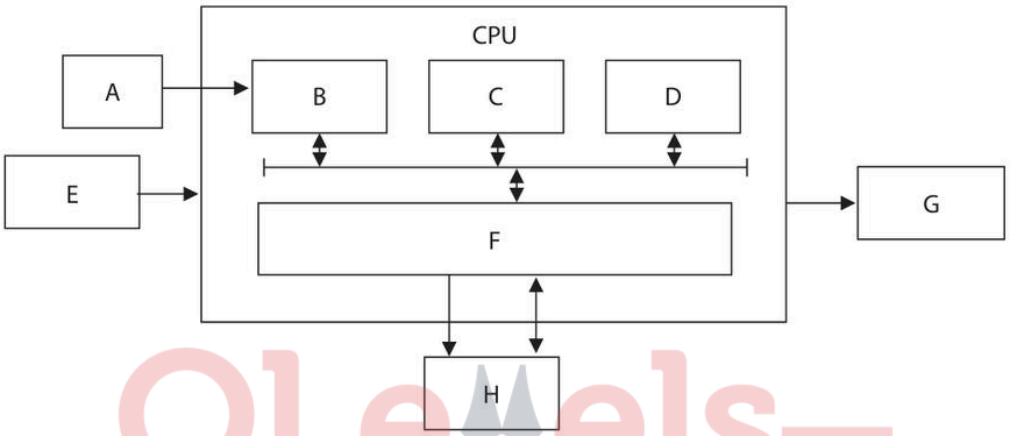


Figure 4

Component	Letter
Control unit	
Output device	
Registers	C
Main memory	
Cache	F
Input device	
Arithmetic/logic unit (ALU)	
Clock	A

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

6. Question 5 · part a · 5 marks

2021 NOV · Paper 01

Further practice · Attempt part a only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

(1)

- ☒ **A** Memory delete signal
- ☒ **B** Memory read signal
- ☒ **C** Memory store signal
- ☒ **D** Memory write signal

(iii) Clock speed is the number of pulses the CPU's clock generates per second.

State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

(1)

(c) An embedded system forms part of a larger system, device or machine.

(i) Give **two** features of an embedded system.

(2)

1 .....

2 .....

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16

P 6 9 2 9 7 A 0 1 6 2 0

6. Question 5 · part a · 5 marks · Working and answer

Working and answer

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7. Question 1 · part a.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part a.ii only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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P 7 2 5 3 8 A 0 2 2 0

7. Question 1 · part a.ii · 1 mark · Working and answer

Working and answer

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8. Question 1 · part b · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part b only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

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- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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8. Question 1 · part b · 1 mark · Working and answer

Working and answer

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Practice workbook · Buses and the fetch-decode-execute cycle

What the address, data and control buses carry and in which direction, the effect of bus width, how the program counter moves on, and putting the cycle's stages in order.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-04

1. Question 2 · part c.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.ii only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

1 .....

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2 .....

.....

(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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.....

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.....

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1. Question 2 · part c.ii · 2 marks

2025 JUN · Paper 01

Worked-example retry · Attempt part c.ii only.

(c) The CPU performs the fetch-decode-execute cycle.

(i) Give **three** factors that affect the performance of the CPU.

(3)

1 .....

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2 .....

.....

3 .....

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(ii) Describe how the address bus is used during the fetch-decode-execute cycle.

(2)

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(d) Classical computers store data using bits that exist in one of two binary states, 1 or 0.

Quantum computers store data in a superposition of states.

Give **two other** characteristics of quantum computers.

(2)

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2 .....

.....

(Total for Question 2 = 12 marks)

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1. Question 2 · part c.ii · 2 marks · Working and answer

Working and answer

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2. Question 3 · part a · 6 marks

2020 NOV · Paper 01

Further practice · Attempt part a only.

3 The central processing unit (CPU) and other hardware components work together with the operating system to run programs and store data.

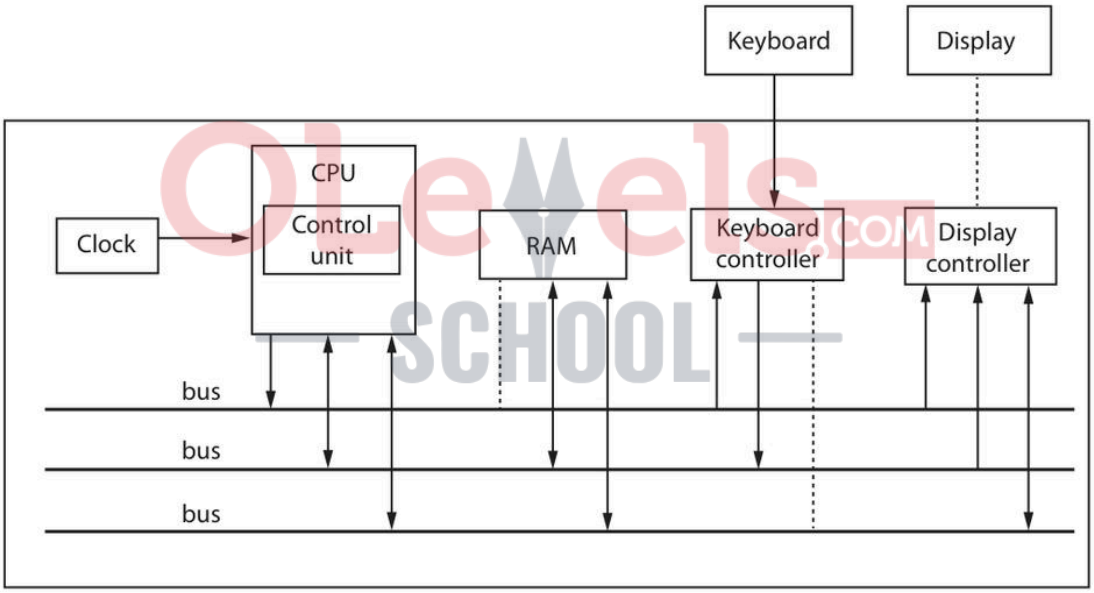
(a) Here is a partially completed diagram showing the hardware components used in the fetch-decode-execute cycle.

The function of the I/O controller is to manage communication between the CPU and input/output devices.

Complete the diagram by:

- labelling the buses
- adding directional arrows to the dotted lines to show the flow of communication.

(6)



(b) The CPU includes an area of cache memory.

Increasing the size of the cache will improve the performance of the CPU.

Explain how increasing the size of the cache improves the performance.

(2)

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2. Question 3 · part a · 6 marks · Working and answer

Working and answer

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3. Question 4 · part d.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.iii only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

(1)

.....

.....

(b) Explain **one** reason why a simulation may require a powerful processor.

(2)

.....

.....

.....

.....

(c) A scientist uses Boolean logic in programs.

Complete the table to show the results of each operation.

(3)

R	S	W	NOT S	R AND W	(NOT S) OR (R AND W)
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1	1	0			
1	1	1			

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DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 1 0 2 0

3. Question 4 · part d.iii · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.iii only.

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(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

(2)

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

(2)

1 .....

2 .....

(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

(2)

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

(2)

Change made	Effect
Increasing the width of the data bus	
Increasing the width of the address bus	

(Total for Question 4 = 14 marks)

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3. Question 4 · part d.iii · 2 marks · Working and answer

Working and answer

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4. Question 4 · part d.iv · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.iv only.

4 Scientists often use models and simulations.

(a) Give **one** reason, other than cost, why scientists use simulations.

(1)

(b) Explain **one** reason why a simulation may require a powerful processor.

(2)

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Complete the table to show the results of each operation.

(3)

R	S	W	NOT S	R AND W	(NOT S) OR (R AND W)
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

P 6 6 4 8 9 A 0 1 0 2 0

4. Question 4 · part d.iv · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part d.iv only.

DO NOT WRITE IN THIS AREA

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(d) Von Neumann developed the stored program concept.

(i) Describe the **stored program concept**.

(2)

(ii) The fetch-decode-execute cycle is the cycle the central processing unit (CPU) follows in order to process instructions.

Name **two** registers used in the cycle.

(2)

1 .....

2 .....

(iii) Describe the role of the address bus and the data bus during the **fetch** part of the cycle.

(2)

(iv) Complete the table to show the effect on a computer system of increasing the width of a bus.

(2)

Change made	Effect
Increasing the width of the data bus	
Increasing the width of the address bus	

(Total for Question 4 = 14 marks)

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4. Question 4 · part d.iv · 2 marks · Working and answer

Working and answer

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5. Question 5 · part b.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

5 Computers can be general-purpose or specific-purpose machines.

(a) The central processing unit (CPU) is responsible for executing a program.

Figure 4 shows some of the hardware components of a computer system.

Complete the table by matching the component to the letters shown in **Figure 4**.

(5)

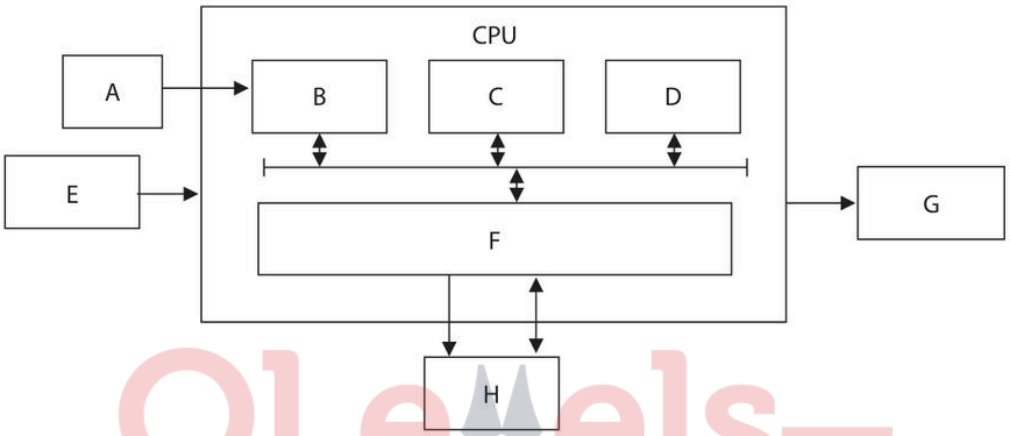


Figure 4

Component	Letter
Control unit	
Output device	
Registers	C
Main memory	
Cache	F
Input device	
Arithmetic/logic unit (ALU)	
Clock	A

DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA
DO NOT WRITE IN THIS AREA

5. Question 5 · part b.i · 2 marks

2021 NOV · Paper 01

Further practice · Attempt part b.i only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

(1)

☒ **A** Memory delete signal

☒ **B** Memory read signal

☒ **C** Memory store signal

☒ **D** Memory write signal

(iii) Clock speed is the number of pulses the CPU's clock generates per second.

State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

(1)

(c) An embedded system forms part of a larger system, device or machine.

(i) Give **two** features of an embedded system.

(2)

1 .....

2 .....

DO NOT WRITE IN THIS AREA

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P 6 9 2 9 7 A 0 1 6 2 0

5. Question 5 · part b.i · 2 marks · Working and answer

Working and answer

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6. Question 5 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

5 Computers can be general-purpose or specific-purpose machines.

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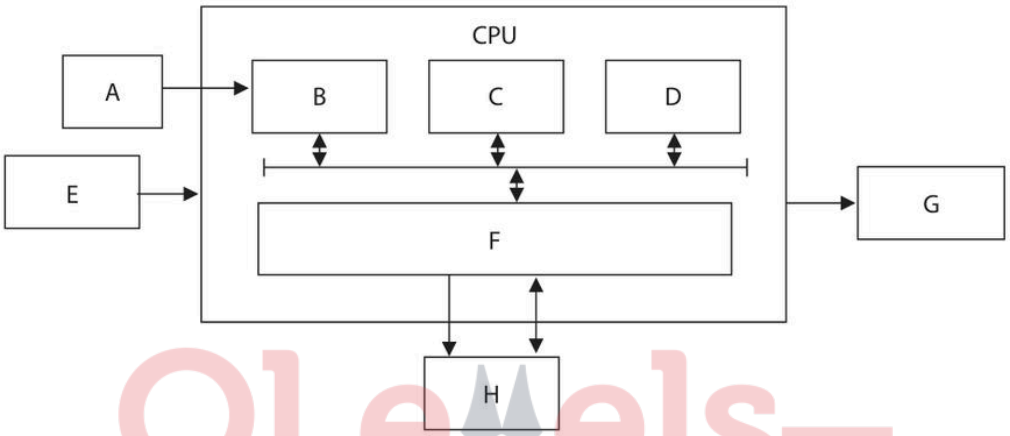


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Component	Letter
Control unit	
Output device	
Registers	C
Main memory	
Cache	F
Input device	
Arithmetic/logic unit (ALU)	
Clock	A

DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA DO NOT WRITE IN THIS AREA

6. Question 5 · part b.ii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.ii only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

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Identify the signal that would be used during the **fetch** stage of the cycle.

(1)

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DO NOT WRITE IN THIS AREA

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P 6 9 2 9 7 A 0 1 6 2 0

6. Question 5 · part b.ii · 1 mark · Working and answer

Working and answer

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7. Question 2 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

2 Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

(1)

.....

.....

(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

(3)

1 .....

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2 .....

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3 .....

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7. Question 2 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

(b) Computers use different types of storage.

(i) Explain **one** reason why computers need both RAM and ROM.

(2)

(ii) Describe how data is read from a magnetic hard drive.

(3)

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(c) Computers perform the fetch-decode-execute cycle.

Complete the table by writing the numbers 1 to 4 in the sequence number column to put each stage in the correct order.

Stage	Sequence number
The instruction is placed onto the data bus	
The Arithmetic and Logic Unit (ALU) performs a calculation	
The address of the next instruction is placed onto the address bus	
The instruction is decoded by the control unit	

(4)

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P 7 5 7 3 6 A 0 7 2 8

7

Turn over

7. Question 2 · part c · 4 marks

2024 JUN · Paper 01

Further practice · Attempt part c only.

(d) Identify which **one** of these buses is unidirectional.

(1)

- ☒ **A** Address bus
- ☒ **B** Control bus
- ☒ **C** System bus
- ☒ **D** Data bus

(Total for Question 2 = 14 marks)

DO NOT WRITE IN THIS

Working and answer

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8. Question 2 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.

2 Computer systems use hardware and software.

(a) Two types of software are application software and system software.

(i) State the purpose of application software.

(1)

.....

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(ii) One function of an operating system is to provide a user interface.

State **three** other functions of an operating system.

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1 .....

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8. Question 2 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.

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OLevels.COM SCHOOL

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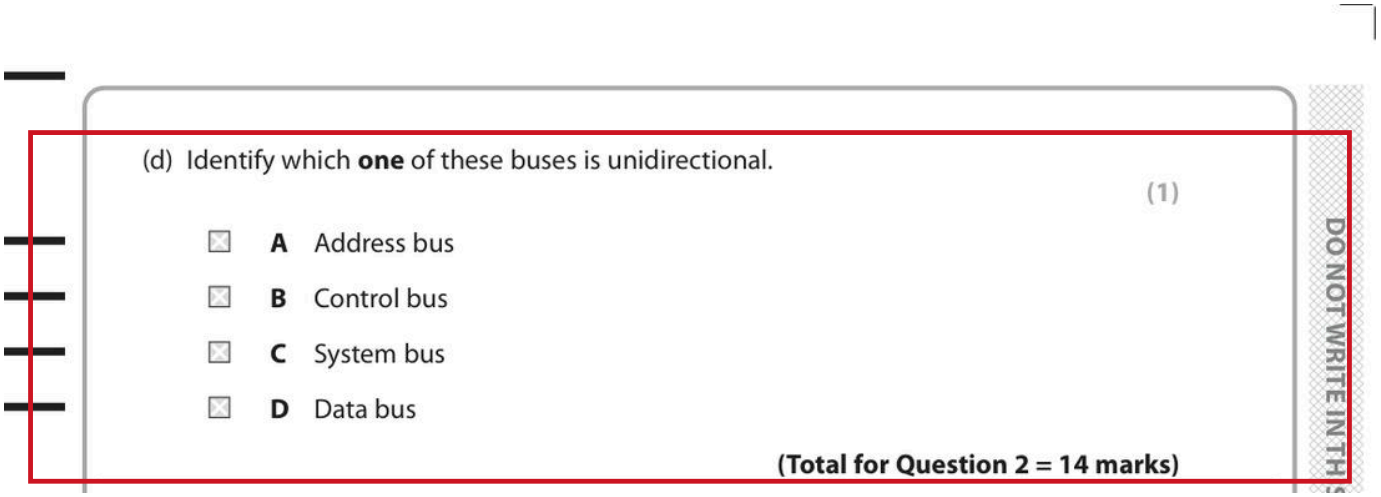
Stage	Sequence number
The instruction is placed onto the data bus	
The Arithmetic and Logic Unit (ALU) performs a calculation	
The address of the next instruction is placed onto the address bus	
The instruction is decoded by the control unit	

(4)

8. Question 2 · part d · 1 mark

2024 JUN · Paper 01

Further practice · Attempt part d only.



(d) Identify which **one** of these buses is unidirectional.

(1)

- ☒ **A** Address bus
- ☒ **B** Control bus
- ☒ **C** System bus
- ☒ **D** Data bus

(Total for Question 2 = 14 marks)

DO NOT WRITE IN TH

Working and answer

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Practice workbook · What makes a CPU faster: clock speed, cores and cache

Clock speed, number of cores, cache size/type and their effect on how fast instructions are processed, including benefits and drawbacks of a higher clock speed and why cache helps.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: CSC-T4-06

1. Question 5 · part b.iii · 2 marks

2023 JUN · Paper 01

Worked-example retry · Attempt part b.iii only.

5 Hardware devices execute programs to carry out a variety of tasks.

(a) A program controls a bee character in an animation.

The bee can turn to face North, East, South or West.

The bee can move any number of steps in the direction it is facing.

Complete the table to show one input and one output required to move the bee.

(2)

Requirement	Example
Input	
Process	Calculate the path the bee will move along to its new position
Output	

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1. Question 5 · part b.iii · 2 marks

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Worked-example retry · Attempt part b.iii only.

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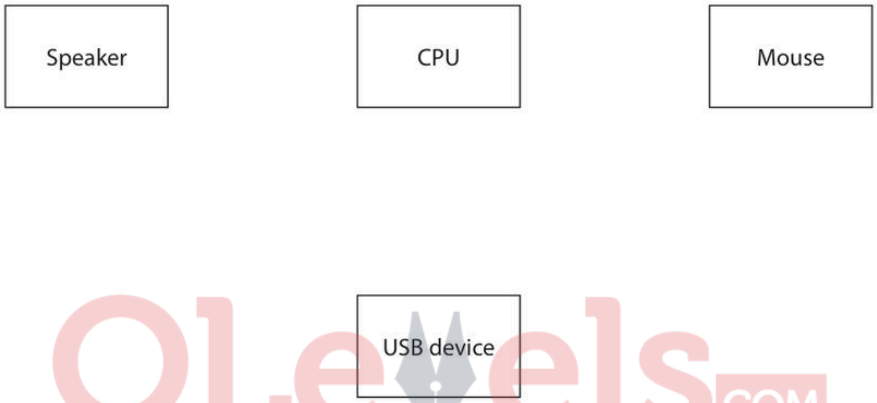
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(b) Computers are made up of hardware components.

(i) Complete the diagram by adding directional arrows between the components to show the flow of communication.

(3)



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1. Question 5 · part b.iii · 2 marks

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Worked-example retry · Attempt part b.iii only.

(ii) Identify what is stored in ROM.

(1)

- ☒ **A** The software firewall
- ☒ **B** The basic input output system
- ☒ **C** The operating system
- ☒ **D** The user interface code

(iii) Cache is used as temporary storage.

One type of cache is located between main memory and the CPU.

Explain **one** reason cache is used in a computer.

(2)

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P 7 2 9 3 7 A 0 1 6 2 4

1. Question 5 · part b.iii · 2 marks

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Worked-example retry · Attempt part b.iii only.

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(c) A washing machine uses several different embedded systems.

One embedded system uses a switch to identify the type of wash cycle selected by the user.

Describe **one other** example of an embedded system found in a washing machine.

(2)

(d) Programs use logic statements to control physical hardware.

(i) A window shuts when the temperature is too cool or it is a rainy night.

The values are defined as:

- A shows it is night time
- B shows it is too cool
- C shows it is raining.

Complete the truth table to show the results of each operation.

Two rows have been done for you.

(3)

A	B	C	A AND C	(A AND C) OR B
0	0	0	0	0
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1	1	1

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1. Question 5 · part b.iii · 2 marks · Working and answer

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2. Question 3 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

3 The central processing unit (CPU) and other hardware components work together with the operating system to run programs and store data.

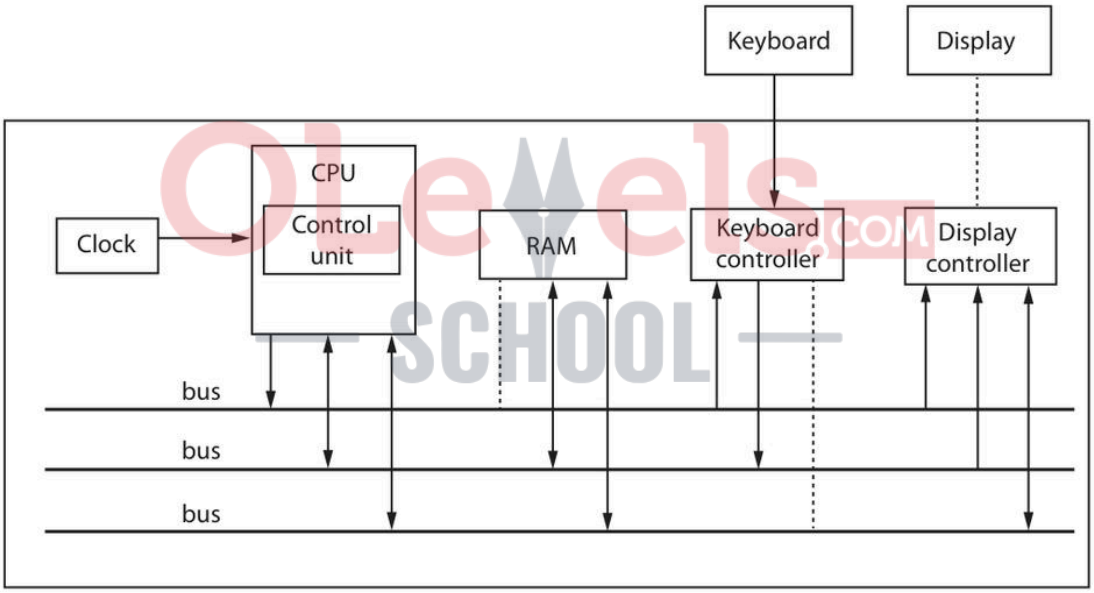
(a) Here is a partially completed diagram showing the hardware components used in the fetch-decode-execute cycle.

The function of the I/O controller is to manage communication between the CPU and input/output devices.

Complete the diagram by:

- labelling the buses
- adding directional arrows to the dotted lines to show the flow of communication.

(6)



(b) The CPU includes an area of cache memory.

Increasing the size of the cache will improve the performance of the CPU.

Explain how increasing the size of the cache improves the performance.

(2)

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2. Question 3 · part b · 2 marks

2020 NOV · Paper 01

Further practice · Attempt part b only.

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(c) Some computers have more than one CPU. The CPUs work in parallel to execute instructions.

Complete the table to show how two CPUs working in parallel could execute the expression:

(5 * 2) + (4 * 3) – (24 / 4)

(4)

CPU 1	CPU 2
5 * 2 = 10	

(d) One function of an operating system is to manage files on secondary storage.

Describe how an operating system manages the storage of a file on random access secondary storage.

(4)

(Total for Question 3 = 16 marks)

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2. Question 3 · part b · 2 marks · Working and answer

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3. Question 3 · part c · 2 marks

2021 JUN · Paper 01

Further practice · Attempt part c only.

3 Zakir wants to buy a new laptop.

(a) He knows that a laptop has both hardware and software.

Explain why the laptop needs both hardware and software.

(2)

(b) Zakir needs to choose between a solid state drive and a hard disc drive.

(i) Give **two** reasons, other than cost and capacity, why Zakir may choose a solid state drive rather than a hard disc drive.

(2)

1

2

(ii) Solid state drives store data on flash memory, which is similar to RAM.

Explain **one** difference between flash memory and RAM.

(2)

(c) One of the laptops has 4 megabytes (MB) of cache memory.

Explain the purpose of cache memory.

(2)

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3. Question 3 · part c · 2 marks

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Further practice · Attempt part c only.

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(d) Zakir uses many devices.

Complete the table by adding **one** tick (✓) in each row to show whether the device is an input device, an output device or both.

(2)

Device	Input	Output	Both
Headset			
Mouse			
Printer			
Touch screen			

(e) Zakir needs to protect his laptop against malware.

Give **two** types of anti-malware software he could install.

(2)

1 .....

2 .....

(Total for Question 3 = 12 marks)

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4. Question 5 · part b.iii · 1 mark

2021 NOV · Paper 01

Further practice · Attempt part b.iii only.

5 Computers can be general-purpose or specific-purpose machines.

(a) The central processing unit (CPU) is responsible for executing a program.

Figure 4 shows some of the hardware components of a computer system.

Complete the table by matching the component to the letters shown in **Figure 4**.

(5)

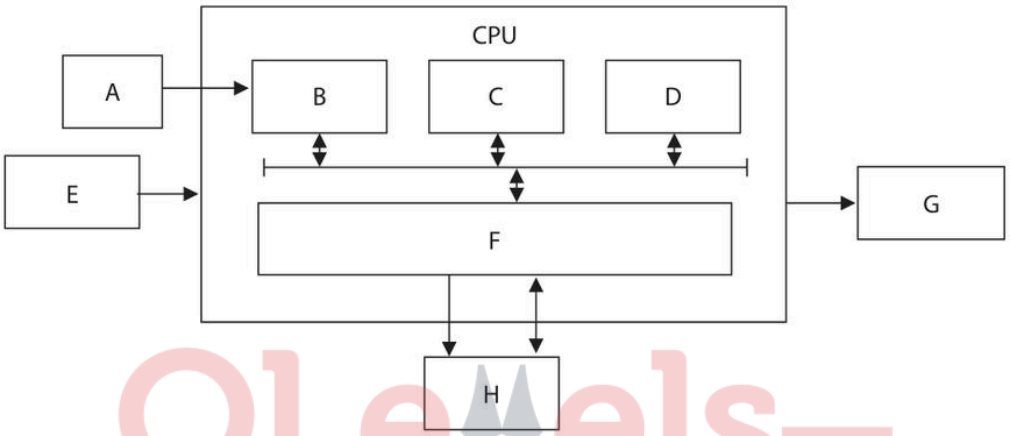


Figure 4

Component	Letter
Control unit	
Output device	
Registers	C
Main memory	
Cache	F
Input device	
Arithmetic/logic unit (ALU)	
Clock	A

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4. Question 5 · part b.iii · 1 mark

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Further practice · Attempt part b.iii only.

(b) The CPU runs a fetch-decode-execute cycle.

(i) Some registers within the CPU are general-purpose whilst others perform a specific function.

One register is the program counter. Its function is to keep track of the next instruction to be fetched.

Describe how it does this.

(2)

(ii) The control bus carries signals from the processor to other components.

Identify the signal that would be used during the **fetch** stage of the cycle.

(1)

- ☒ **A** Memory delete signal
- ☒ **B** Memory read signal
- ☒ **C** Memory store signal
- ☒ **D** Memory write signal

(iii) Clock speed is the number of pulses the CPU's clock generates per second.

State how increasing the speed of the clock impacts on the fetch-decode-execute cycle.

(1)

(c) An embedded system forms part of a larger system, device or machine.

(i) Give **two** features of an embedded system.

(2)

1

2

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16

P 6 9 2 9 7 A 0 1 6 2 0

4. Question 5 · part b.iii · 1 mark · Working and answer

Working and answer

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5. Question 1 · part c.i · 1 mark

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Further practice · Attempt part c.i only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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P 7 2 5 3 8 A 0 2 2 0

5. Question 1 · part c.i · 1 mark · Working and answer

Working and answer

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6. Question 1 · part c.ii · 1 mark

2022 JUN · Paper 01

Further practice · Attempt part c.ii only.

1 Computer systems have both hardware and software components.

(a) The central processing unit (CPU) uses the fetch-decode-execute cycle.

(i) State what is meant by the term **program instruction**.

(1)

(ii) State what is meant by the term **memory address**.

(1)

(b) Identify the component of the CPU that provides temporary data storage.

(1)

- ☒ **A** Address bus
- ☒ **B** Data bus
- ☒ **C** Control unit
- ☒ **D** Register

(c) The performance of the CPU is affected by the clock speed.

(i) Give **one** benefit of having a higher clock speed.

(1)

(ii) Give **one** drawback of having a higher clock speed.

(1)

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6. Question 1 · part c.ii · 1 mark

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Further practice · Attempt part c.ii only.

(d) Identify which **one** of these describes a sequential computational model.

(1)

- ☒ **A** Program instructions are read one after another from external storage
- ☒ **B** Program instructions are executed by multiple agents working together
- ☒ **C** Program instructions are executed in parallel by different cores
- ☒ **D** Program instructions are executed one after another

(e) A program can be written in a high-level or a low-level language.

(i) Give **one** reason for writing a program in a low-level language.

(1)

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(ii) State the purpose of an assembler.

(1)

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(iii) Complete the table by adding **one** tick (✓) in **each** row to match the description.

(3)

Description	Compiler	Interpreter
Translates the program each time it is executed		
Produces permanent object code		
Translates line by line		
Translates the whole program before it is run		
Generates a list of errors once the complete program has been translated		

(Total for Question 1 = 11 marks)

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6. Question 1 · part c.ii · 1 mark · Working and answer

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7. Question 5 · part b · 2 marks

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Further practice · Attempt part b only.

5 A team is developing a computer game.

(a) Complete the table by adding a tick (✓) to show the characteristics of program language translators.

You may select more than one translator per characteristic.

(3)

Characteristic	Translator: Assembler	Translator: Interpreter	Translator: Compiler
Does **not** have to be present in memory to execute the translated code.			
Translates high level language source code.			
Executes high level code until either the program has finished, or a syntax error is encountered.			
Translates low level language source code into machine code.			

(b) Luca is testing the computer game.

The computer that he is using is an older model with a processor that runs at a low clock speed.

Explain **one** drawback of testing the game using this computer.

(2)

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7. Question 5 · part b · 2 marks

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(c) Anna uses utility software.

Complete the table by adding the type of utility software she should use to solve each issue.

Issue	Utility software
Her word processor cannot read all of a file because it is corrupt.	
She has a JPEG image and needs to send it to a publisher who can only read PNG image files.	
There has been a flood and files on a server have been damaged, but Anna needs to access her data.	
Her hard drive is running very slowly when she tries to access a file.	

(4)

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7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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8. Question 2 · part c.i · 3 marks

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Further practice · Attempt part c.i only.

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2 Computer systems use both hardware and software.

(a) Give **two** output devices found on smartphones.

(2)

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2 .....

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(b) Computer systems use both primary and secondary storage.

(i) Identify which **one** of these uses storage space on the hard drive as additional memory.

(1)

- ☒ **A** Cache memory
- ☒ **B** Random-access memory
- ☒ **C** Read-only memory
- ☒ **D** Virtual memory

(ii) Computer systems can use cloud storage.

Describe what is meant by **cloud storage**.

(2)

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8. Question 2 · part c.i · 3 marks

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Further practice · Attempt part c.i only.

(c) The CPU performs the fetch-decode-execute cycle.

(i) Give **three** factors that affect the performance of the CPU.

(3)

1 .....

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2 .....

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3 .....

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(ii) Describe how the address bus is used during the fetch-decode-execute cycle.

(2)

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(d) Classical computers store data using bits that exist in one of two binary states, 1 or 0.

Quantum computers store data in a superposition of states.

Give **two other** characteristics of quantum computers.

(2)

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2 .....

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(Total for Question 2 = 12 marks)

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8. Question 2 · part c.i · 3 marks · Working and answer

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